Executive brief: profit-aware retention targeting

To: Head of Customer Retention | **From:** Retention analytics team | **Decision requested:** approve a 90-day retention pilot

The issue. Churn puts about **$468,468** of 12-month customer margin at risk across the 7,043 customers of the sample company (IBM's fictional US telco; USD). Today's rule spends the budget on the customers most likely to leave, ignoring what they are worth and what an offer costs.

Recommendation

1. **Target by expected value, not by risk alone.** Make an offer only where churn risk x customer value x the offer's *save rate* (the share of would-be leavers it keeps) exceeds its cost. Here that means **fibre customers on month-to-month contracts with high bills**: 100% of the recommended list have fibre and 97% a monthly contract, with a median bill of $89 against $70 overall. Low-risk and low-bill customers are left out.
2. **Match the offer to the segment:** $35 credit for moving to a 1-year contract (150), 6 months free TechSupport + OnlineSecurity (150), One-time $20 bill credit (24).
3. **Test before scaling:** one randomised pilot wave of 1,620 customers, half held out, and scale up only if the measured save rate clears that list's **13.8% break-even**.

Why: the same budget returns more

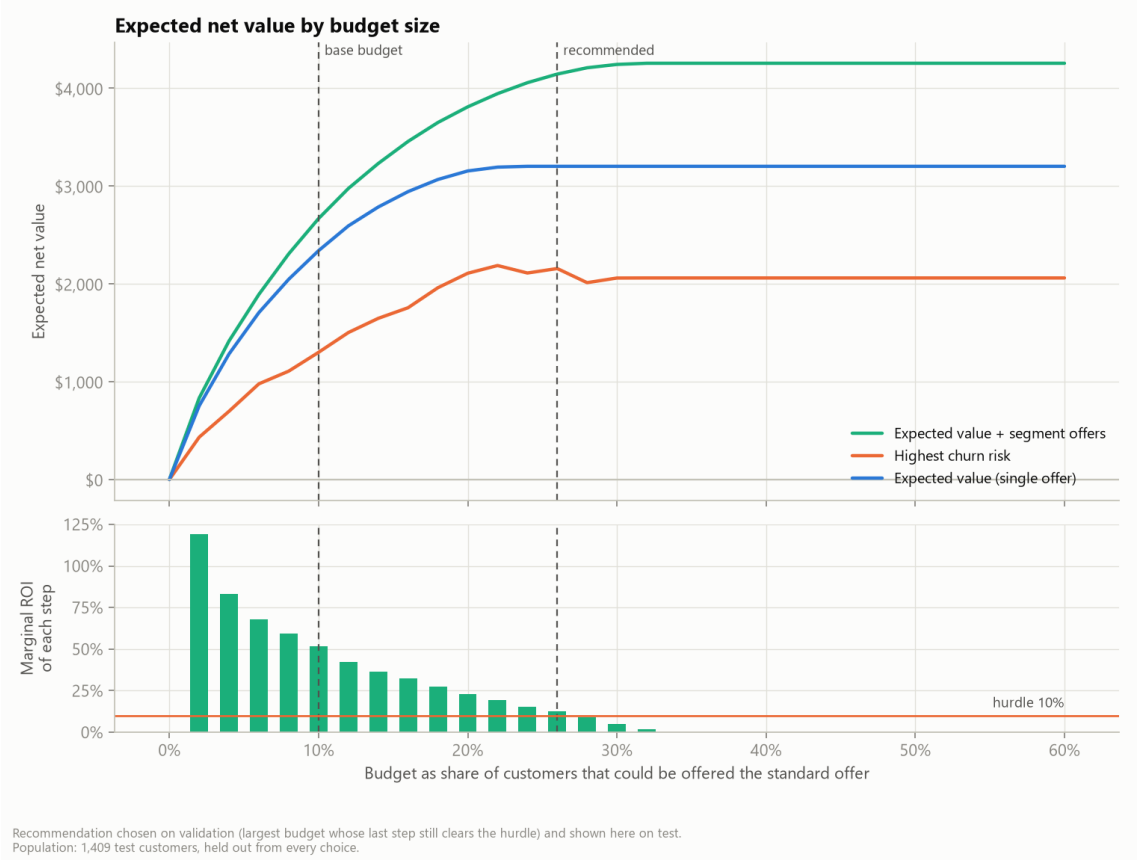
1,409 test customers (held out from every choice), fixed $3,500 budget:

Option	Customers contacted	Expected net value	ROI	Precision among contacted	High-risk/high-value coverage
Do nothing	0	$0	n/a	n/a	0%
Call highest churn risk (today)	140	$1,301	37%	71%	12%
Call highest value	140	-$1,037	-30%	29%	30%
Expected value, one standard offer	140	$2,340	67%	60%	96%
Expected value + segment offers (recommended)	**153**	**$2,668**	**76%**	**60%**	**98%**

- **The recommended rule is expected to add $1,368 for the same spend**: $1,039 from choosing customers by expected value and $328 from matching offers. Valued with actual churn (at the assumed save rates), the gain is $1,077 (95% interval $229 to $2,002); the part from matching offers is not proven.
- Risk-only is more precise but spends on risky customers whose value cannot repay an offer. Calling the most valuable customers loses money: risk and value are needed together.

How much to spend

Each extra budget step returns less (figure). Keeping only steps that still return at least 10% (decided on validation data) gives about **$45,737 for the whole base** ($6.49 per customer). On the test customers this buys 324 offers for $9,150: expected net value $4,145, ROI 45%. Beyond 26% of customers nobody has positive expected value.



How sure are we?

- **Who to target is robust.** The model ranks churners well (ROC-AUC 0.83) and its probabilities are within 1.9 points of observed churn. With all assumptions drawn at once, expected-value targeting beats risk-only in 100% of draws.
- **How much an offer saves is not.** Lists break even at a save rate of 11.3% (base budget) and 13.8% (recommended); we assume 20%. Only 46% of churners left for price or a competitor's offer, so 20% means keeping 43% of those an offer can reach.
- **Downside:** the recommended list loses money in 17% of draws, and in the worst 5% it loses more than $12,217 across the base; the smaller base-budget list loses in 9%.

Risks and guardrails

- **No causal evidence yet.** Save rates are assumptions and contact can even raise churn, so the pilot keeps a random control group.
- **Sample data.** Real operators lose far fewer customers than this sample's 27%. At 15% average churn only 7% of customers would be worth this offer, so offers and contact must get cheaper (for example SMS); retrain on our own data (Vietnam deployment note).
- **Fairness.** Seniors are contacted 3.3x as often as others (2.6x under today's rule), 1.7x after adjusting for risk. Age and gender are not model inputs; Legal/DPO reviews each wave.
- **Compliance.** Consent, Do-Not-Call checks, contact limits and a data-protection impact assessment.

Decisions requested

1. **Approve one pilot wave:** 810 customers offered, 810 held out; offer and contact cost about $22,875, expected net value of the offered half $10,363, worst case (no one saved) -$22,875.
2. **Name owners:** Retention (offers), Call centre (capacity), Finance (margin, offer cost), Legal/DPO (consent, DPIA).
3. **Agree the rule now:** after 90 days, **go** if the effect is statistically significant and at least 13.8%; **stop** if clearly below; otherwise extend, at most 2 more waves. At a true save rate of 20% the rule says go 95% of the time. Decision about 19 weeks after approval.